

Shiloh Oni

Nigeria • shilohadebisi@gmail.com • +234 915 102 9763 • [Google Scholar](#) • [GitHub](#)

EDUCATION

Federal University of Agriculture, Abeokuta (FUNAAB)

Doctor of Veterinary Medicine (D.V.M)

April 2017 – Sep 2024

- CGPA: 3.71/5.0

RESEARCH INTERESTS

Natural Language Processing (NLP), Multilingual NLP, Information Retrieval, Low-Resource Speech and Language Technologies, Speech and Audio Processing, Speech Recognition and Synthesis, Computational Neuroscience, AI for Healthcare, Computer Vision, Medical Imaging.

PUBLICATIONS

- **Adekoya, T. O.** and **Oni, S. A.** 2026. “AfriVTON-Bench: Evaluating the Robustness of Virtual Try-On Models to African Textile Patterns and Garment Structures.” *Accepted for Oral Presentation, Women in Machine Learning (WiML) Workshop at the International Conference on Machine Learning (ICML) 2026*, Seoul, South Korea.
- Abdulmatin Omotoso, Habeeb Shopeju, Adejumobi Joshua, and **Shiloh Oni**. 2025. “Improving BGE-M3 Multilingual Dense Embeddings for Nigerian Low-Resource Languages.” In *Proceedings of the 9th Widening NLP Workshop*, pages 224–229, Suzhou, China. Association for Computational Linguistics.

EXPERIENCE

Veterinary Doctor (NYSC)

Government Veterinary Clinic, Warri, Delta State

Warri, Nigeria

June 2025 – May 2026

- Delivered primary clinical care across small animals and livestock, including diagnosis, surgical procedures, vaccination, and post-operative monitoring under the state veterinary service.
- Supported disease surveillance and public-health reporting on zoonotic conditions, contributing to the clinic’s case records and outbreak response notes.
- Counselling livestock farmers and pet owners on preventive care, nutrition, and biosecurity practices.

Data Science Intern

SeqHub Analytics

Remote

June 2024 – Dec 2024

- Built an AI-powered chatbot for Sickle Cell patients using LLMs and RAG, improving response accuracy by 70%.
- Analyzed user interaction data to enhance personalization and optimize medical information delivery.

Independent Researcher

ML Collective

Remote

Sep 2023 – Present

- Lead and contribute to applied ML research projects spanning multilingual information retrieval, few-shot medical imaging, and multimodal deepfake defence (see Projects section for details).
- Two co-authored works awarded Best Poster at Deep Learning Indaba 2025, Kigali.
- Participate actively in weekly MLC reading sessions, reviewing cutting-edge machine learning papers and supporting members with essay reviews and conference preparation materials.

ML Engineer Intern

Future Academy Africa

Remote

Sept 2023 – Sep 2024

- Implemented TensorFlow and OpenCV models for real-time queue prediction and optimization.
- Built a Chrome extension automating company data extraction for internal data teams.

Data Analyst

MopCare

Remote

Sept 2022 – Aug 2023

- Developed data-driven tools for caregiver–client matching, increasing satisfaction and service quality by 25%.
- Automated validation systems that reduced caregiver mismatch rates by 20%.

- Designed and deployed a scalable Python-based recommender system using Scikit-learn and collaborative filtering, boosting recommendation accuracy by 15% and user engagement by 20%.
- Developed an ARIMA time-series forecasting model, improving demand planning accuracy by 25%, cutting inventory costs by 12%, and enhancing supply chain efficiency.

PROJECTS

AfriVTON-Bench: Robustness of Virtual Try-On Models on African Garments 2025 – 2026
Co-Lead Researcher, with Testimony Adekoya

- Designed and constructed a benchmark of 111 African garment images across 16 categories and 7 countries, paired with a 50-image VITON-HD Western control set, yielding 1,012 person–garment evaluation pairs.
- Audited VITON-HD and DressCode for African textile and skin-tone representation gaps, motivating the benchmark and its evaluation protocol.
- Evaluated FASHN-VTON and LEFFA across the benchmark using gLPIPS, pLPIPS, gSSIM, and DISTS, and proposed *tonal drift* as a new perceptual metric capturing skin-tone shift between input and synthesized output.
- Surfaced failure modes specific to high-frequency wax-print patterns and non-Western draping silhouettes (pattern dissolution, drape collapse, tonal drift). Paper accepted for oral presentation at WiML @ ICML 2026.

BGE-M3 Multilingual Dense Embeddings for Nigerian Low-Resource Languages 2024 – 2025
Lead Researcher, ML Collective

- Fine-tuned the BGE-M3 dense embedding model on three Nigerian languages for information retrieval over local document corpora, addressing the multilingual coverage gap in standard retrieval encoders.
- Achieved a 20% improvement in Mean Reciprocal Rank (MRR) over the baseline BGE-M3 across the three target languages.
- Published in *Proceedings of the 9th Widening NLP Workshop* (Suzhou, 2025); awarded Best Poster at Deep Learning Indaba 2025.

Few-Shot Learning for Multi-Disease Blood Smear Diagnosis 2024 – 2025
Lead Researcher, ML Collective

- Built a few-shot classifier for sickle cell anemia, malaria, leukemia, and trypanosomiasis from limited blood smear images, targeting diagnostic settings where labelled data is scarce.
- Deployed the trained model for real-time inference on unseen blood smear samples with class-level prediction output.
- Awarded Best Poster at Deep Learning Indaba 2025.

Multimodal Deepfake Defence & Anti-Spoofing 2024 – Present
Co-Investigator, ML Collective

- Contributing to a multimodal detection pipeline that combines visual (face) and audio (voice) cues to flag deep-faked and spoofed media in low-resource African contexts.
- Working on dataset curation, baseline detector benchmarking, and cross-modal fusion under compression and adversarial perturbation.
- Participating in weekly research sync and peer review across a global volunteer team spanning over ten countries.

PRESENTATIONS AND CONFERENCES

Presenter & Attendee **Accra, Ghana**
African Computer Vision Summer School (ACVSS) 2026 *July 2026*

- Presenting research on robustness of virtual try-on models to African textile patterns and garment structures; engaging in technical sessions on computer vision research across the African continent.

Oral Presenter **Seoul, South Korea**
Women in Machine Learning (WiML) Workshop, ICML 2026 *2026*

- Oral presentation of “AfriVTON-Bench: Evaluating the Robustness of Virtual Try-On Models to African Textile Patterns and Garment Structures” (co-authored with Testimony Adekoya).

Poster Presenter **Kigali, Rwanda**
Deep Learning Indaba 2025 *Aug 2025*

- Presented two posters: “Improving BGE-M3 Multilingual Dense Embeddings for Nigerian Low-Resource Languages” and “Learning From Less: Few-Shot Learning for Multi-Disease Blood Smear Diagnosis”.

Attendee

Data Science Africa (DSA) 2025 Conference

University of Ibadan, Nigeria

June 2025

- Engaged in sessions on applied AI research and community-driven data science in Africa.

Attendee

PyCon Africa

Accra, Ghana

Sept 2024

- Participated in sessions on open-source software, AI ethics, and Python community development.

Volunteer & Attendee

Deep Learning Indaba

Accra, Ghana

Sept 2023

- Attended talks, workshops, and networking sessions on machine learning research and applications across Africa; supported event registration as a volunteer.

TEACHING EXPERIENCE

Data Science Tutor

AI Invasion, FUNAAB

Abeokuta, Nigeria

April 2022 – Sept 2022

- Tutored 15 students remotely on data science and machine learning fundamentals including NumPy, pandas, and exploratory data analysis.

Data Science Tutor

SheCode Africa

Abeokuta, Nigeria

June 2022 – Aug 2022

- Mentored women across Africa entering technology careers, supporting their progression from foundational data science skills to internship- and job-ready competence.

SKILLS

- **Programming & Tools:** Python, SQL, Bash, Git, LaTeX, Jupyter, Docker.
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, scikit-learn, HuggingFace Transformers, Diffusers, PEFT/LoRA, fine-tuning, few-shot learning.
- **Natural Language Processing:** Multilingual NLP, dense and sparse retrieval, RAG, embedding models (BGE-M3, sentence-transformers), tokenization, evaluation (MRR, nDCG).
- **Computer Vision & Medical Imaging:** OpenCV, image classification, segmentation, virtual try-on benchmarking, perceptual metrics (LPIPS, SSIM, DISTS), blood smear analysis.
- **Speech & Audio:** Transformer architectures for speech, audio feature extraction, anti-spoofing pipelines.
- **Information Retrieval & Vector Search:** FAISS, ChromaDB, hybrid retrieval, reranking.
- **Data Analysis & Visualization:** pandas, NumPy, Matplotlib, Seaborn, statistical analysis, time-series (ARIMA).
- **Veterinary & Clinical:** Clinical diagnosis, small-animal and livestock medicine, surgery, disease surveillance, zoonotic case reporting, biosecurity counselling.
- **Soft Skills:** Research writing, peer review, technical mentoring, cross-disciplinary collaboration, public speaking.

SCHOLARSHIPS AND AWARDS

- **Best Poster Awards**, Deep Learning Indaba 2025, Kigali, Rwanda — for “Improving BGE-M3 Multilingual Dense Embeddings for Nigerian Low-Resource Languages” and “Learning from Less: Few-Shot Learning for Multi-Disease Blood Smear Diagnosis”.
- **Recipient**, Women Techsters Fellowship (2025) — selected for the competitive fellowship empowering women in technology across Africa.
- **AWS AI & ML Scholarship** for the Udacity AI Programming with Python Nanodegree.
- **Certificate of Best Reviewer**, Deep Learning Indaba 2025 — recognized for exceptional contribution and feedback quality in the paper and application review process.
- **Certificate of Excellence and Leadership**, College of Veterinary Medicine (COLVET), FUNAAB — honored for exemplary academic performance and leadership.